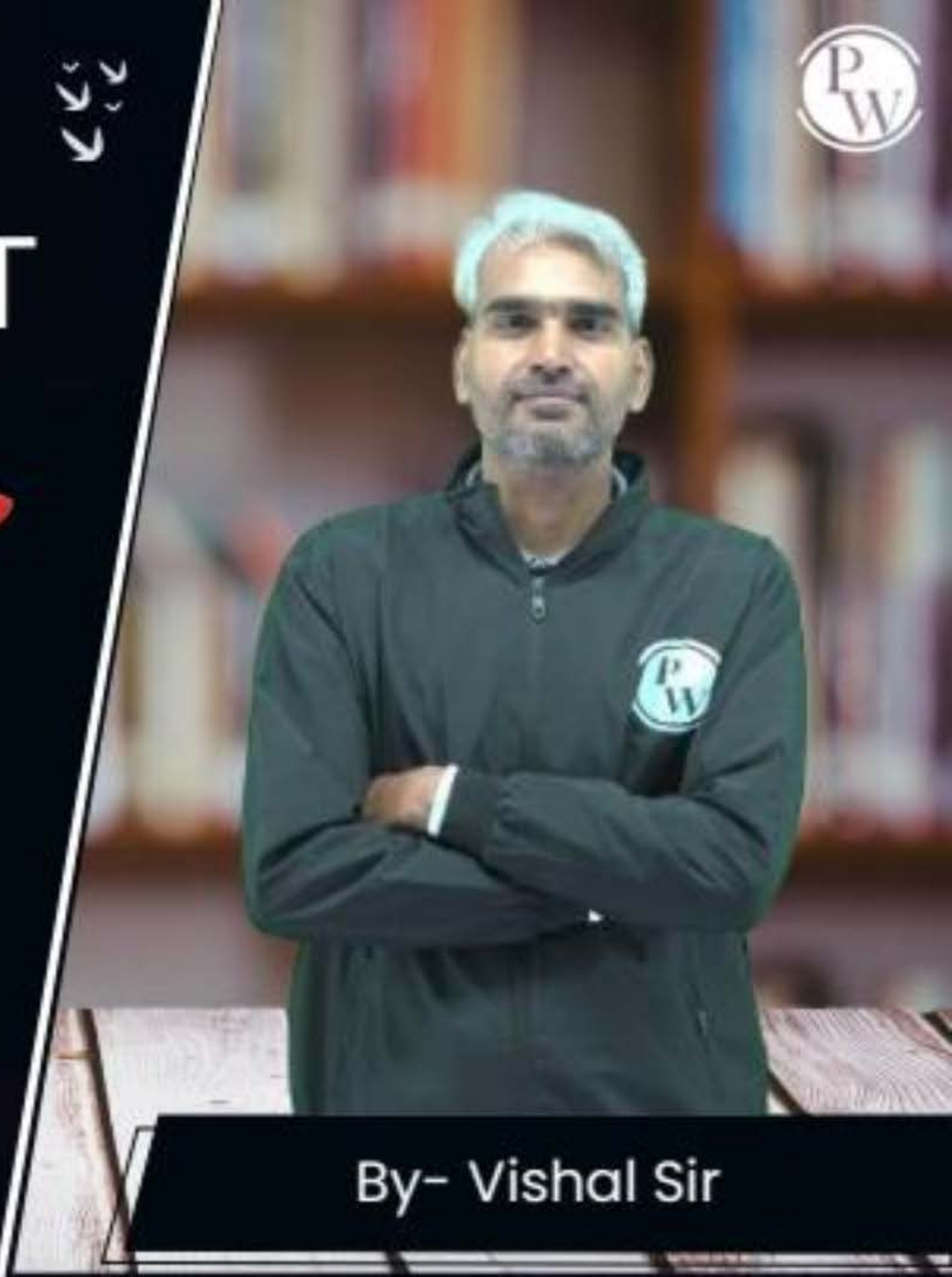
Computer Science & IT



Database Management System

File organization And Indexing

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



- **Topic** — Time stamp ordering protocol with Thomas write rule
- **Topic** — Wait-die protocol for deadlock prevention
- **Topic** — Wound-wait protocol for deadlock prevention
- **Topic** — Database, File and Records

Slide

Topics to be Covered



- Topic: Blocking factor
- Topic: Organization of Records
- Topic: IO cost without index
- Topic: Index file
- Topic: Search key and type of files
- Topic: IO cost with index file

File organization & Indexing

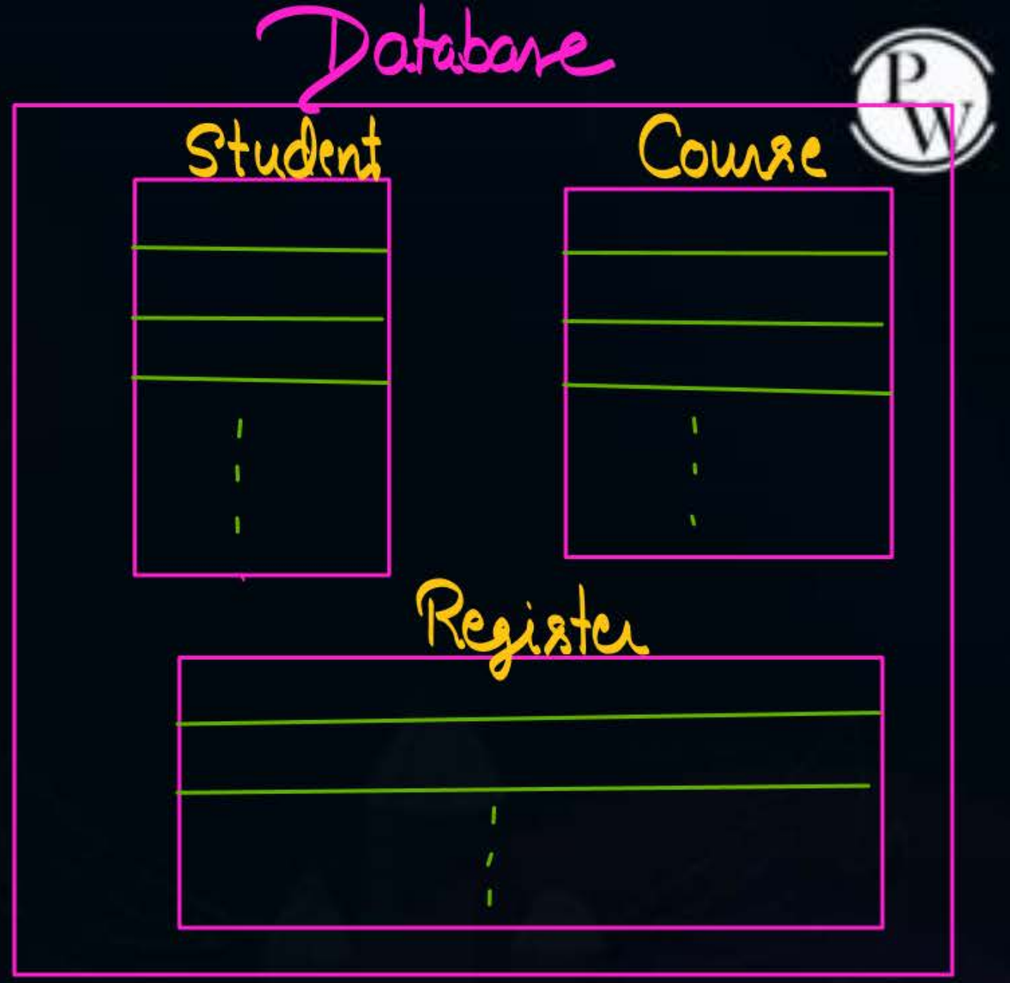


Topic : Database-File-Records

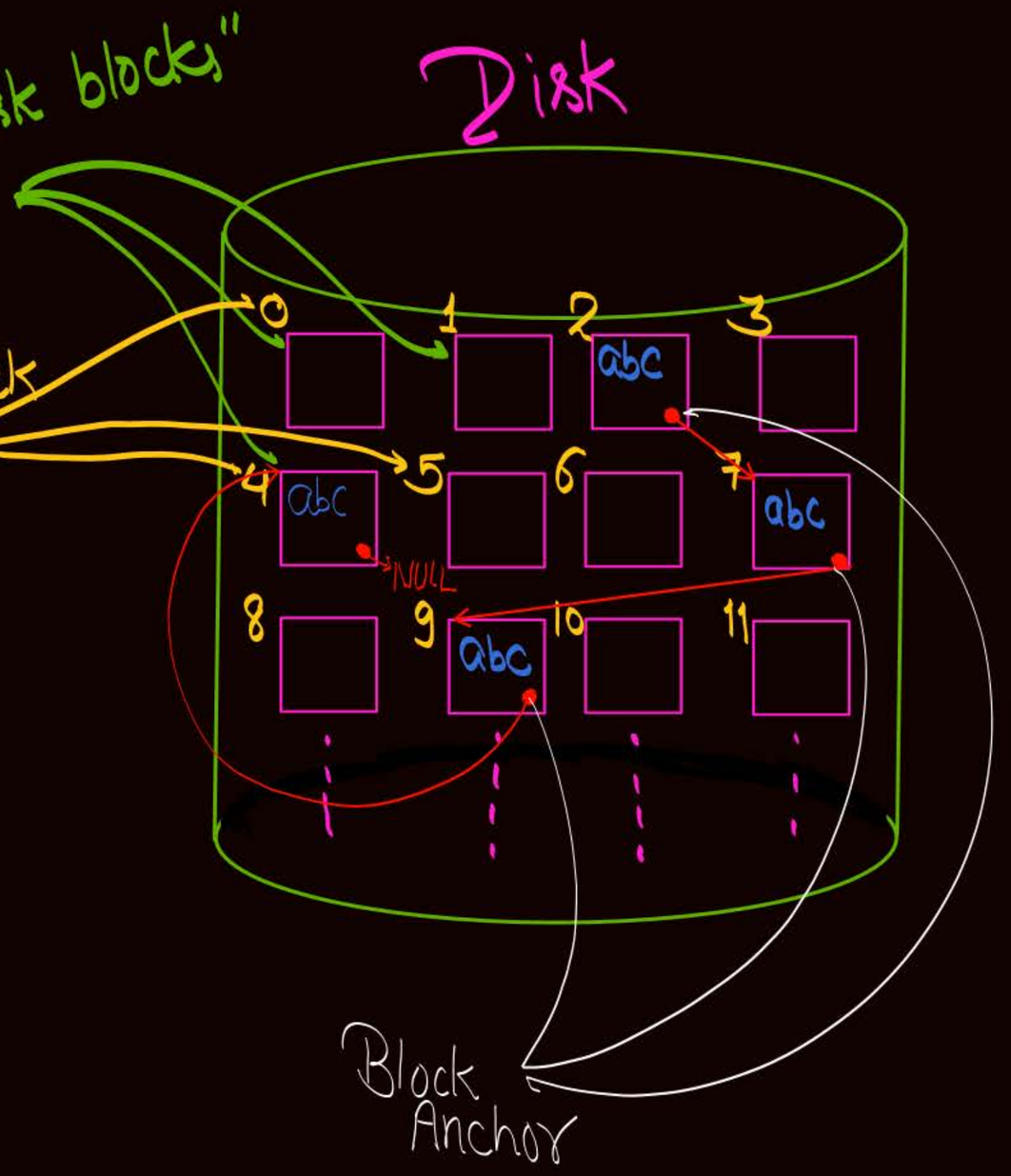
* Database is a collection of files
* Files are used to store the records.
* Disk blocks are used to store the records of the file

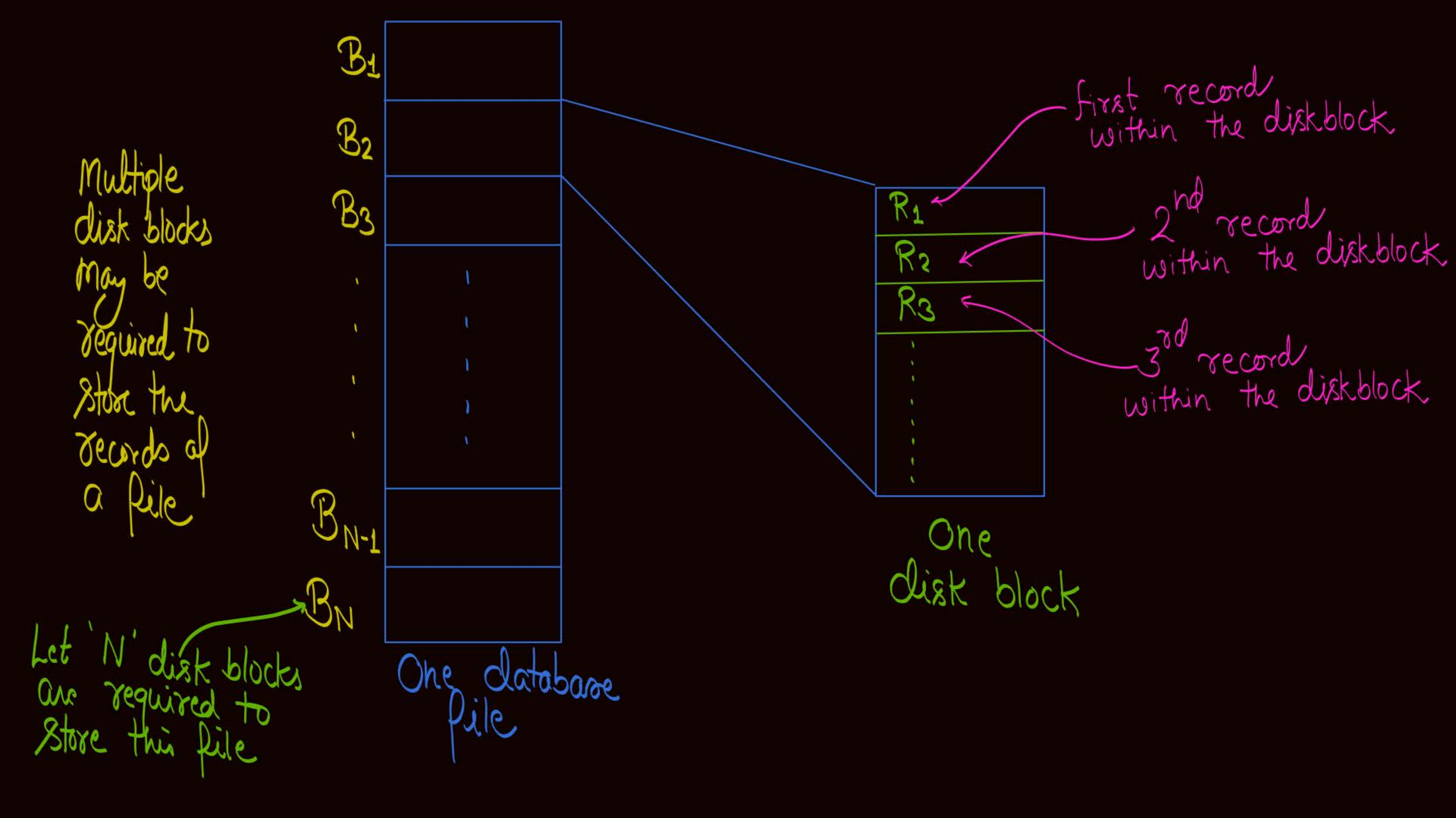
One disk block can be used to store the records of a single file.

└ Records of different files can not be stored in the same disk block



- Each disk block within the disk will be of same size
- Each block of the disk is identified by a unique number known as "disk block address"
- Each disk block can store multiple records of the same file (not from different files)




Multiple disk blocks may be required to store the records of a file
B_1
B_2
B_3
B_{N-1}
B_N
Let 'N' disk blocks are required to store this file
One database file
R_1
R_2
R_3
First record within the disk block
2nd record within the disk block
3rd record within the disk block
One disk block



Topic : Types of Records

* Two types of records are possible in file system

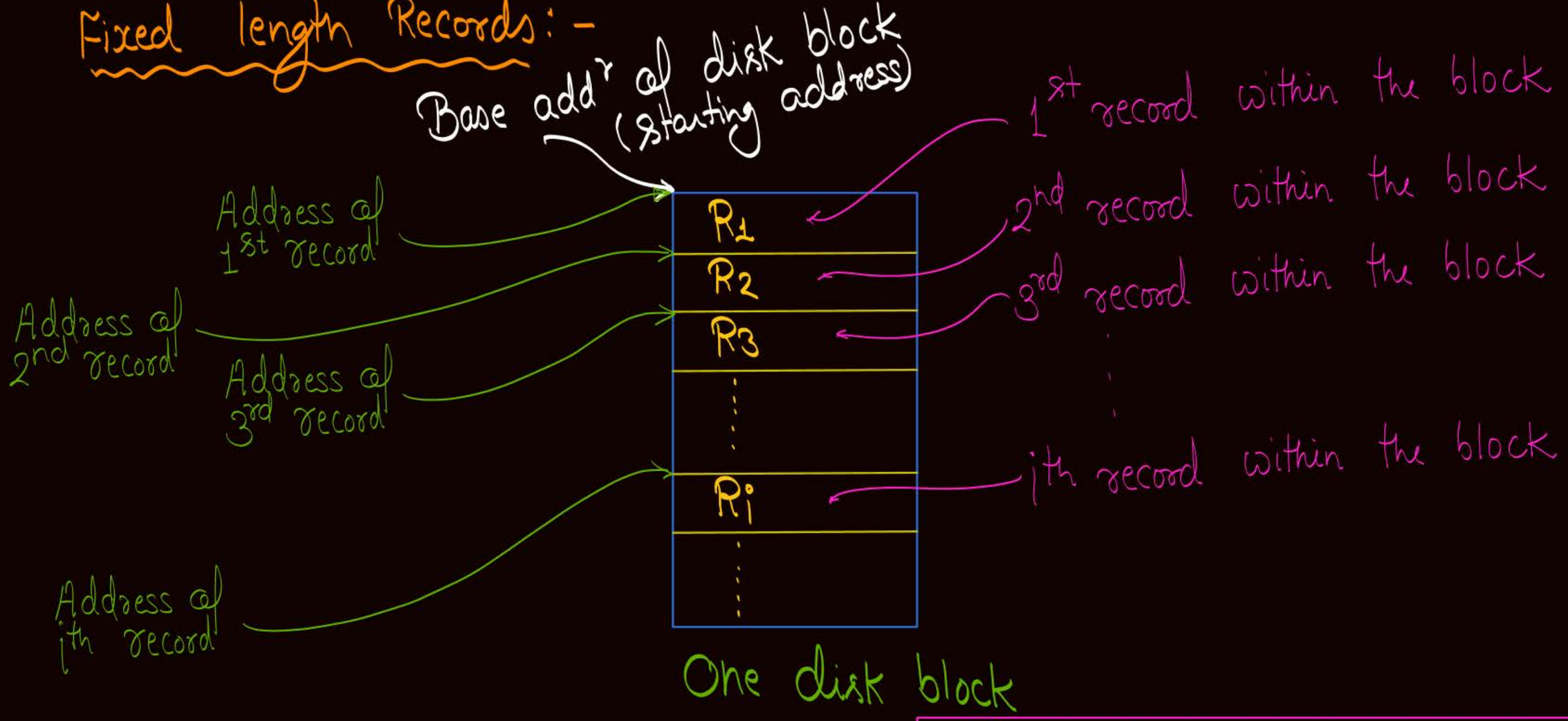
1. Fixed length records
2. Variable length records.



Topic : Fixed length records

When each record of the file is of same size

Fixed length Records :-



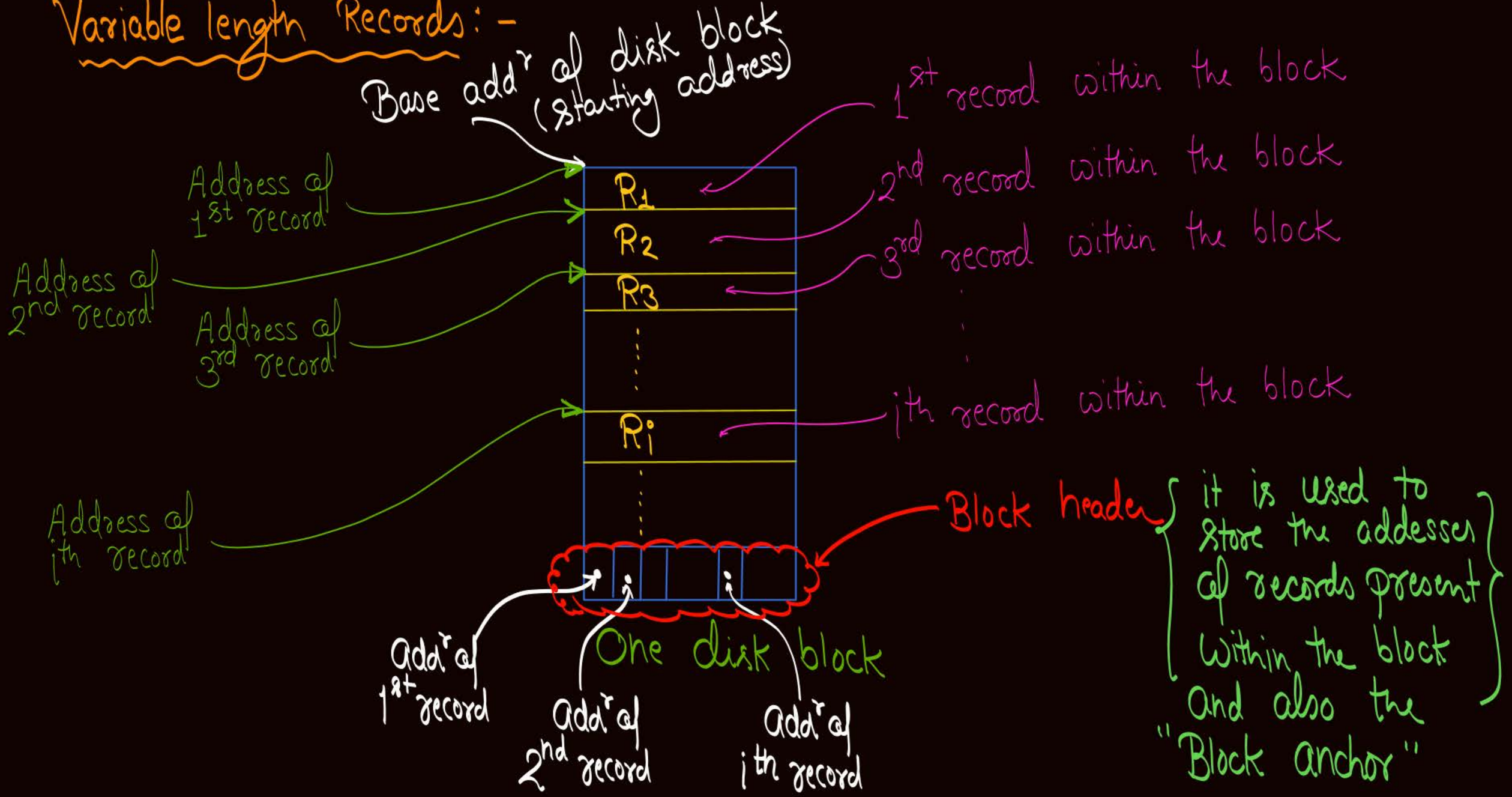
$$\text{Address of } i^{th} \text{ record within the block} = \text{Base add}^{r} \text{ of the block} + (i-1) * \text{Size of each record}$$



Topic : Variable length records

When records of the same file may be of different sizes.

Variable length Records :-



"Block anchor" :- Block anchor gives the address of the next block of the disk used to store the records of the file.

* Note :- Block header may be required with fixed length records as well, in order to store the address of the next disk block of the file {i.e; to store block anchor}

Note:→ ① In the questions if nothing is given about block header size and/or block anchor size then consider it as "Zero", i.e., the complete disk block size with be used to store the records of the file.

② In the question, If block header size is given, then Effective space available in the disk block to store the records of the file will be

= (Disk block size − Block header size)



Topic : Blocking Factor

"Bf"

- Blocking factor of database block is defined as no. of records that can be stored in one disk block.

$$\text{Blocking factor} = \left(\frac{\text{Disk block size}}{\text{Record size}}\right)$$

When block header size is not given

$$\text{Blocking factor} = \left(\frac{\text{Disk block size} - \text{Block header size}}{\text{Record size}}\right)$$

When Block-header size is given



Topic : IO cost

* IO Cost of a record access can be defined as number of disk blocks that needs to be transferred from secondary memory to main memory in order to access that record



Topic : Organization of Records

* There are two ways in which records of the file can be organized within the disk block

✓ (1) Unspanned organization → { Every record must be stored Completely within the same block }

✓ (2) Spanned organization → { Some of the records may span in two blocks }



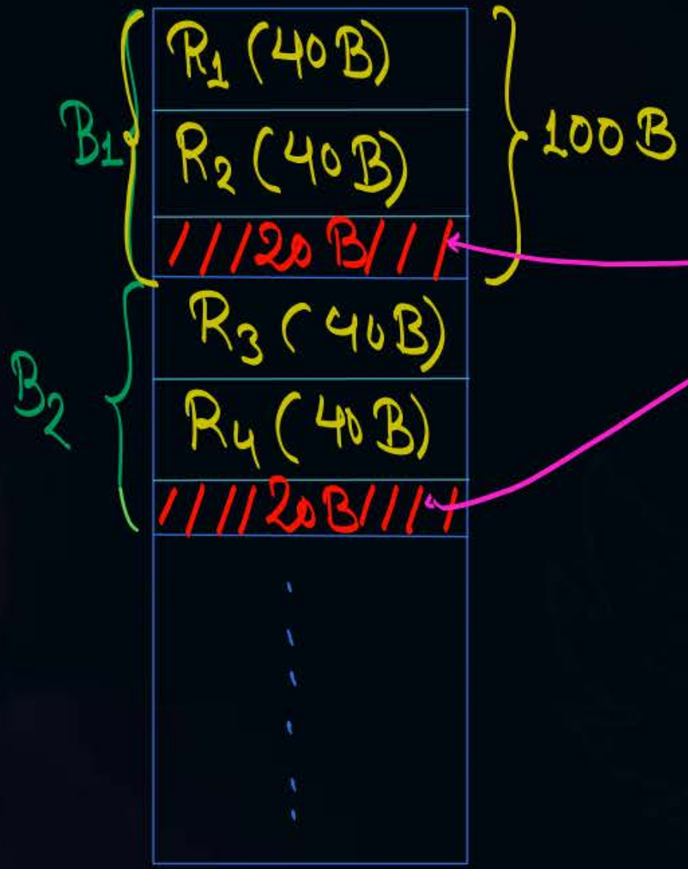
Topic : Un-spanned Organization

{ When a record of the file is not allowed to span in more than one block }

Consider,

Disk block size = 100 Bytes

Record size = 40 Bytes



Available space is not sufficient to store a record completely, therefore this space is wasted

↓

and it is called "internal fragmentation"

$$\text{Blocking factor} = \left\lfloor \frac{(\text{Disk block size} - \text{Block header size})}{\text{Record size}} \right\rfloor$$

$$\text{Blocking factor} = \left\lfloor \frac{100}{40} \right\rfloor = 2$$

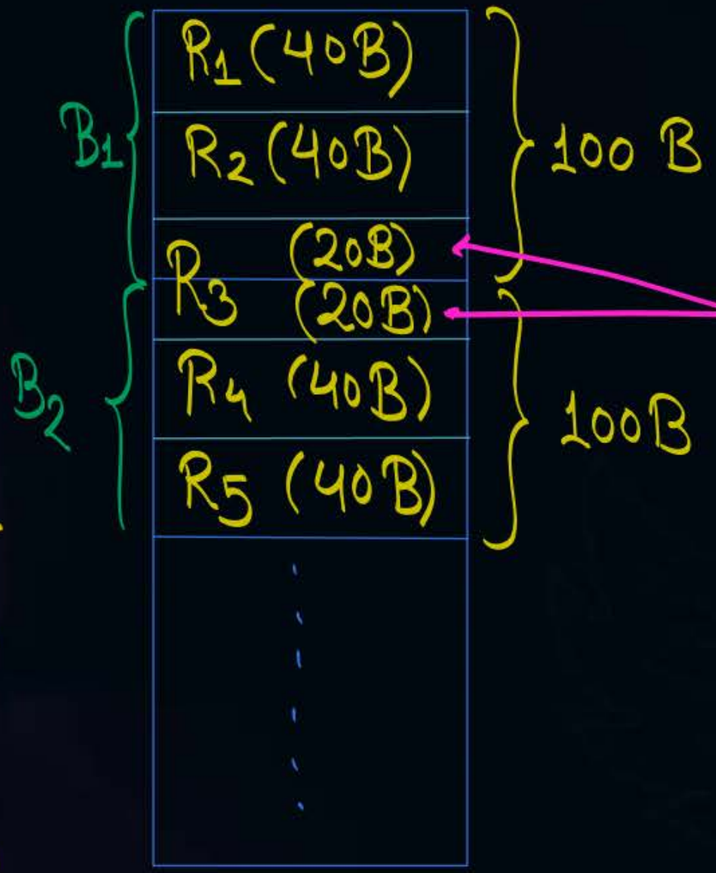


Topic : spanned Organization

Consider,

Disk block size = 100 Bytes

Record size = 40 Bytes



No internal fragmentation {Except for the last block used to store the records of the file}

$$\text{Blocking factor} = \frac{\text{Block size} - \text{Block header size}}{\text{Record size}}$$

$$= \frac{(100-0)}{40} = 2.5$$



Topic : Search-key & Type of File

Need not have unique value

Search key :- Attribute / field used to search for a record in a database file is called search key

Retrieve Record of the Employee with "Eid = E_6"

⇓

Search for records will be performed based on attribute "Eid"

⇓

∴ Eid will be the search key

Employee database file

Eid	--other attributes---	Dept_id	
E_3		2	B_1
E_1		1	
E_4		2	B_2
E_6		3	
E_2		4	B_3
E_5		1	
⋮	⋮	⋮	

Retrieve records of all the Employee with dept_id = 2

⇓

Search for records will be performed based on attribute "dept_id"

⇓

∴ "dept_id" will be the search key



Topic : Search-key & Type of File

Need not have unique value

Search key :- Attribute / field used to search for a record in a database file is called search key

- **Types of file :-** There are two types of files

  (1) **Unordered file (Heap file) :-** If records of the file are not Physically ordered {i.e. not sorted} based on value of search key attribute, then file is called unordered file

  (2) **ordered file :-** If records of the file are Physically ordered (sorted) based on the values of search key attribute, then file is called ordered file.

Retrieve Record of the Employee with "Eid = E_6"

⇓

Search for records will be performed based on attribute "Eid"

⇓

∴ Eid will be the search key

⇓

Records of the file are not ordered w.r.t. attribute "Eid"

∴ w.r.t. "Eid" as search key, file is unordered

Employee database file

Eid	--other attributes---	Dept_id	
E_3		2	B_1
E_1		1	
E_4		2	B_2
E_6		3	
E_2		4	B_3
E_5		1	
⋮	⋮	⋮	

Retrieve records of all the Employee with dept_id = 2

⇓

Search for records will be performed based on attribute "dept_id"

⇓

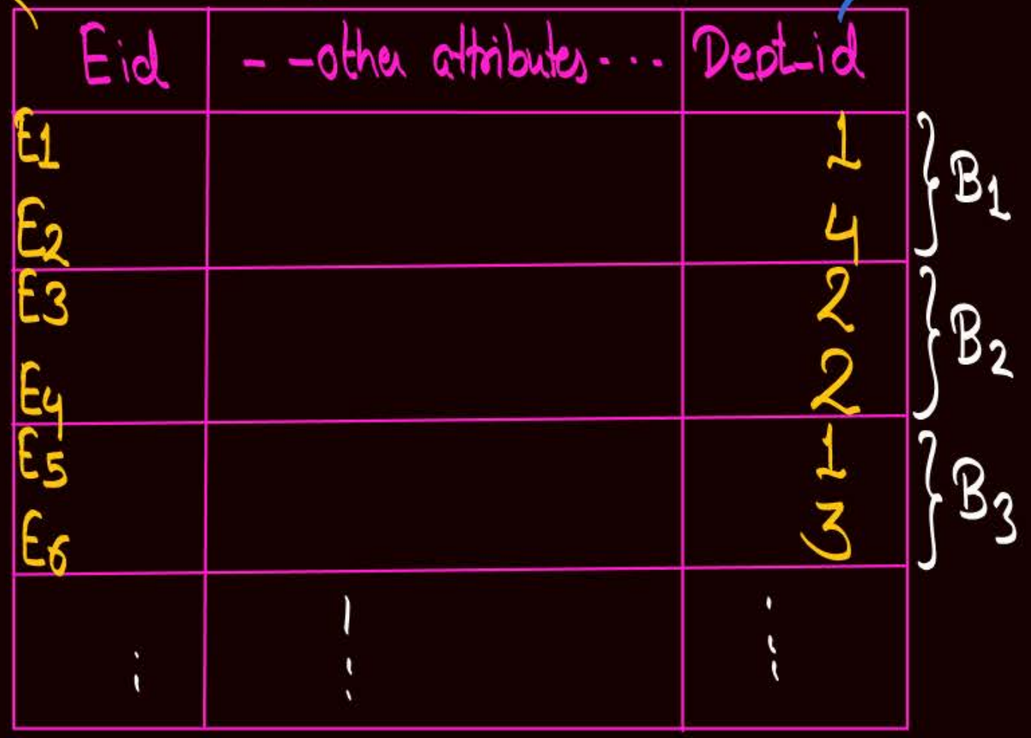
∴ "dept_id" will be the search key

⇓

Records of the file are not ordered w.r.t. "dept_id"

∴ w.r.t. "dept_id" as search key file is unordered

Employee database file



- Retrieve Record of the Employee of Eid = E6

⇓

Search will be performed based on Eid,
∴ Eid is the search key

⇓

* Record of the file are Physically ordered based on search key (i.e. Eid)
∴ file is ordered file w.r.t "Eid" as search key

Records are still Unordered w.r.t "dept_id"

⇓

∴ If search key is the dept_id, then file will be Unordered file

Note:→ A database file can be ordered w.r.t. at most one attribute, at a time.

0 or 1

- 0 → unordered w.r.t. each attribute
- 1 → ordered w.r.t exactly '1' attribute and unordered w.r.t. other attributes.



Topic : IO cost without indexing

* Let "N" is the total number of disk blocks required to store the given database file,

(i) Worst case IO Cost (When file is un-ordered) $= N$ { Search will be linear search }

(ii) Worst Case IO Cost (When file is ordered) $= \lceil \log_2 N \rceil$ { If file is ordered, then we can perform binary search }

Note: If database size is too large, then "N" value will also be a very large value and in that case $\lceil \log_2 N \rceil$ will also be significantly large.
∴ We use Concept of Indexing to reduce this IO Cost.

Ordered file Vs Unorder file
{Also known as Heap file}

Ordered file	Unorder file
① Binary search is possible	① Binary search is not possible
② Insertion of a new record in an ordered file will be a costly operation {Because record must be stored at its appropriate Position in the file}	② Insertion of a new record will be easy {we can simply insert the new record in the last block of the file}



Topic : Index file

Index files are used to reduce the IO cost.

✓ Each entry in the index file contains only two fields

* ① Search key attribute value
* ② Pointer pointing to the record / disk block corresponding to the search key value.



2 mins Summary

- Topic: Blocking factor
- Topic: Organization of Records
- Topic: IO cost without index
- Topic: Index file
- Topic: Search key and type of files
- Topic: IO cost with index file

Slide

THANK - YOU